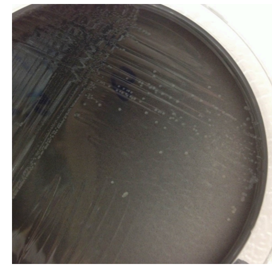
Apr 13, 2026

Extraction of DNA from canine feces for detection of *Campylobacter jejuni* using the MagMax™ CORE Nucleic Acid Purification Kit on KingFisher™ Flex Instrument



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We use this protocol and it's working

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Disclaimer

Reference to any commercial materials, equipment, or process does not in any way constitute approval, endorsement, or recommendation by the Food and Drug Administration.
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Abstract

Campylobacter jejuni is a significant cause of foodborne illness in humans and animals. Rapid detection of animals shedding *C. jejuni* is a key step in limiting the scope of outbreaks and reducing the potential exposure of humans and other animals. Manual DNA extraction methods, although effective, can be labor-intensive and time-consuming. The utilization of automated or semi-automated methods can significantly reduce the time needed for DNA extraction and significantly increase the number of samples that can be processed in a given amount of time. This method was developed to provide a standard method for semi-automated DNA extraction from feces for the detection of *C. jejuni* in veterinary patients.

Validation data (in-house and by an independent laboratory via collaborative study such as Randomized Blinded Method Test) are available upon request.

Image Attribution

Sara Lawhon

Guidelines

All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. All work with specimens that potentially contain *C. jejuni* should follow the best practices outlined in the most current version of Biosafety in Microbiological and Biomedical Laboratories (BMBL) and institutional guidance.

Reporting of positive test results should follow all local, state, and federal regulations.

Follow the MagMax^{TM} CORE Nucleic Acid Purification Kit manufacturer's recommendations, warnings and procedural guidelines.

Materials

For DNA Extraction
1X PBS (Life Technologies cat#10010-023)
Applied Biosystems™ MagMAX™ CORE Nucleic Acid Purification Kit (ThermoFisher Scientific, cat#A32700/A32702)
VetMAX™ Xeno™Internal Positive Control DNA (ThermoFisher cat#s A29762/A29764)
Adhesive PCR Plate Foils, or equivalent (ThermoFisher Scientific, cat# AB0626)

Plasticware - please confirm that plasticware works with your equipment.
KingFisher™ 96 tip comb for DW magnets, 100 combs (ThermoFisher Scientific, cat#97002534)
KingFisher™ Flex Microtiter Deepwell 96 plates (ThermoFisher Scientific, cat#95040460)
KingFisher™ 96 KF microplates (200 µL), (ThermoFisher Scientific, cat#97002540)*
*Note: Only works with KingFisher™ Flex 96 Heating Block (Cat. No. 24075420). Not to be used with KingFisher™ Flex 96 Deep-Well Heating Block (Cat. No. 24075430).

From any manufacturer:
Microcentrifuge tubes - 1.5-mL and 2-mL tubes sterile
50 mL conical tubes - sterile
Pipette tips (1 mL, 200 µL, 20 µL and 10 µL)
Serological pipette, 25 mL

For PCR
TaqMan Fast Virus 1-step Master Mix (ThermoFisher Cat. Nos. 4444432/4444434/4444436)
VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767)
Positive control DNA from *C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560
Negative control DNA from *E. coli* ATCC 25922

	A	B
	Oligo Name	**5′-3′ Sequence**
	gyrA-F	AAGATACGGTCGATTTTGTTCCA
	gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
	gyrA-P	5′-[FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ1]-3′

Primer and probe sequences.

Primers and probe are diluted to a working concentration of 10 µM with a final concentration in the total reaction volume of 0.5 µM.

Equipment needed:

KingFisher™ Flex Magnetic Particle Processor (ThermoFisher Scientific, cat#5400110)
Biotang Inc Microplate Shaker, or equivalent titer plate shaker (for mixing beads with samples; Fisher Scientific™ cat#50-751-4965)
Applied Biosystems™ 7500 Fast Real-Time PCR System or similar real-time PCR machine
Scale for weighing 8 g of feces
Pipettes for appropriate volumes
Centrifuge for microcentrifuge tubes capable of speeds of up to 16,000 X g
Benchtop centrifuge with plate adaptors (for lysate preparation in plates
Laboratory mixer, Vortex or Equivalent
Optional: Qiagen (MoBio) Vortex Adapter (Cat#13000-V1-24)

Reference for primers and probe

Iijima Y, Asako NT, Aihara M, Hayashi K. (2004) Improvement in the detection rate of diarrhoeagenic bacteria in human stool specimens by a rapid real-time PCR assay. J Med Microbiol. 2004 Jul;53(Pt 7):617-22. PMID: 15184531

Protocol materials

VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific Catalog #**A29762/A29764

1X PBS **Life Technologies Catalog #**10010-023

MagMAX™ CORE Nucleic Acid Purification Kit **Thermo Fisher Scientific Catalog #**A32700/A32702

Safety warnings

All manufacturers' intended use, safety information, storage guidelines, and recommendations should be followed.

Appropriate safety and personal protective equipment should be worn while working with chemicals including a suitable lab coat, disposable gloves, and protective eyewear.

Campylobacter jejuni is infectious to humans and is a biological safety level 2 organism. Work with specimens that may contain this organism should follow all applicable institutional safety guidelines.

Background

1 Fecal samples should be stored at 4-8 °C refrigerated until processed.

2 Samples should be **processed within 72 hours of receipt at the laboratory**.

3 If performing bacterial culture, bacterial culture plates and enrichment broth should be incubated in a microaerophilic environment at 42 °C + or - 2 .

4 Remaining samples from the associated culture method can be used for molecular detection.

5 Before using the MagMax^{TM} CORE Nucleic Acid Purification Kit, invert bottles of solutions and buffers to ensure thorough mixing.

6 In this protocol, you will mix samples with reagents from the MagMax^{TM} CORE Nucleic Acid Purification Kit by either using a plate shaker or by pipetting up and down.

If you are using a plate shaker, you will want to determine the maximum setting of your plate shaker prior to running samples.

6.1 First verify that the plate fits securely on the plate shaker.

6.2 Test the maximum setting that you can use on the plate shaker by adding 1 mL of water to each well of a test plate and then covering the plate with sealing foil.

Next, determine the maximum setting that you can use on your shaker without any of the water splashing onto the sealing foil.

7 To prevent cross-contamination, 1) cover the plate or tube strip during the incubation and shaking steps, to prevent spill-over and 2) carefully pipet reagents and samples to avoid splashing.

8 To prevent nuclease contamination, wear laboratory gloves throughout all procedures, use nucleic acid-free pipette tips to handle reagents, and decontaminate the lab benches and pipettes before use.

9 This protocol uses

VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific Catalog** #A29762/A29764

as a control for PCR inhibition.

Note

Note: This reagent will be added to the MagMAX™ CORE Lysis Solution as a DNA extraction control and/or after DNA extraction is complete to serve as the PCR inhibition control. If the laboratory has a system in place for DNA extraction and PCR inhibition, that method can be substituted in this protocol.

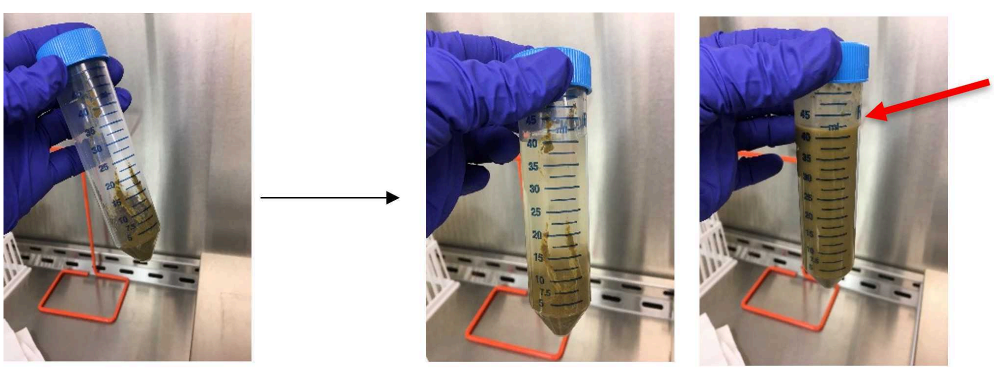
10 Prior to using this protocol, the script for the MagMAX™ CORE kit should be loaded onto the instrument. The script is: **KingFisher Flex heated script: MagMAX CORE_Flex.bdz**

Sample preparation

2m 30s

11 Use 8 g of feces in 50 mL conical tubes

12 Add 32 mL 1X PBS **Life Technologies Catalog** #10010-023 to 8 g of feces. The feces in PBS will hereafter be called the feces-PBS slurry.



8 g feces

Add **32 mL PBS** to 8 g feces to prepare the feces-PBS slurry and allow the tube to sit for 2 minutes.

Vortex slurry for 30 seconds. Use slurry for direct PCR.

When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.

Making the Feces-PBS slurry.

12.1 Allow samples to sit for 00:02:00 after adding the 32 mL PBS 2m

12.2 Vortex samples for 00:00:30 to prepare fecal slurry 30s

PCR Method - DNA Extraction 1m

13 Extract DNA from the feces using the

MagMAX™ CORE Nucleic Acid Purification Kit **Thermo Fisher Scientific Catalog #**A32700/A32702

following the manufacturer's protocol (outlined below).

14 Before processing the fecal slurry, prepare the Processing Plates.

	A	B	C	D	E
	Plate ID	**Plate Position**	**Plate Type**	**Reagent**	**Volume per well**
	Sample Plate	1	Deep Well	*The sample plate will be set up in subsequent protocol steps and will be loaded on the KingFisher™ Flex Instrument at Step 24.	
	Wash Plate 1	2	Deep Well	MagMAX™ CORE Wash Solution 1	500 µL
	Wash Plate 2	3	Deep Well	MagMAX™ CORE Wash Solution 2	500 µL
	Elution	4	Standard	MagMAX™ CORE Elution Buffer	90 µL
	Tip Comb	5	Standard	Place a tip comb in the plate	

Table 1. Plate setup: KingFisher™ Flex instrument

15 Before processing the fecal slurry, prepare the Bead-Proteinase K mix.

15.1 Vortex the MagMAX™ CORE Magnetic Beads thoroughly to ensure that the beads are fully resuspended.

15.2 Combine the following components for the required number of samples plus 10%.

	A	B
	Component	**Volume per sample**
	MagMAX™ CORE Magnetic Beads	20 µL
	MagMAX™ CORE Proteinase K	10 µL
	Total Bead/PK Mix	30 µL

Table 2. Preparation of the Magnetic Beads and Proteinase K

Note

The manufacturer recommends that you prepare new Bead/PK mix for each processing run. If necessary, the Bed PK mix can be stored at 4 °C for up to 1 week.

15.3 Add the Bead/PK Mix to the wells in Sample Plate

16 Prepare the CORE Lysis Solution.

16.1 Combine the following components for the required number of samples plus 10%.

	A	B
	Component	**Volume per sample**
	MagMAX™ CORE Lysis Solution	450 µL
	VetMAX™ Xeno™ Internal Positive Control DNA	4 µL
	Total Lysis Solution and Internal Positive Control DNA	454 µL

Table 2. Preparation of the Lysis Solution

Note

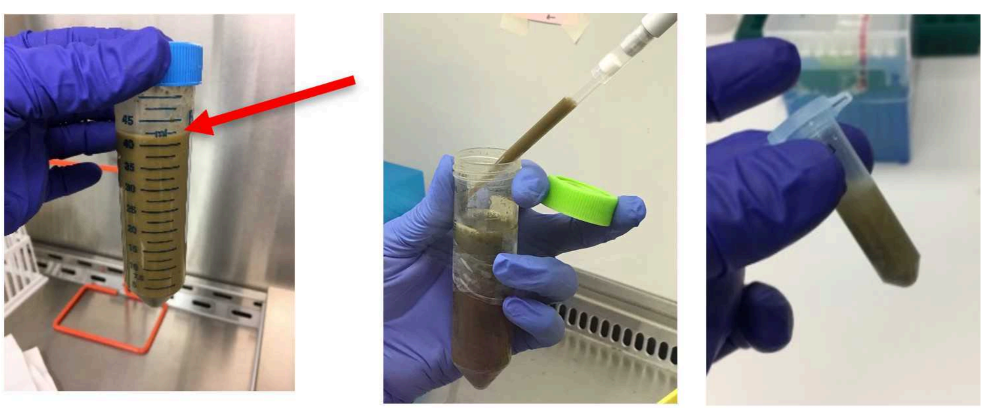
Validation of the protocol used 4 µL VetMAX™ Xeno™ Internal Positive Control DNA per reaction instead of the 2 µL as noted in the manufacturer's protocol for consistency with prior manual extraction methods. Laboratories may choose to validate the use of 2 µL VetMAX™ Xeno™ Internal Positive Control DNA per reaction internally.

16.2 Mix by inverting the tube or bottle at least 10 times.

Note

(Optional) Store Lysis Solution at room temperature for up to 24 hours.

17 To a microcentrifuge tube, add 1 mL of feces-PBS slurry from Sample Preparation Step 12.



When transferring sample from the feces-PBS slurry pipette from the meniscus at the red arrow.

Pipetting the Feces-PBS slurry to a microcentrifuge tube.

18 Centrifuge the sample at 100 x g, Room temperature, 00:01:00 1m

19 Samples can be processed either in tubes (Step 20) or plates (Step 21)

20 Processing in tubes

20.1 Add 454 µL of the Lysis Solution to a new 2-mL tube.

20.2 Transfer 200 µL of the supernatant of the centrifuged fecal slurry to the tube.

20.3 Vortex vigorously for 00:03:00 or until the sample is suspended. 3m

Note

Optional - The sample in the microcentrifuge tube can be placed in a vortexer with Qiagen (MoBio) Vortex Adapter (Cat#13000-V1-24)

20.4 Centrifuge the sample at 15000 x g, Room temperature, 00:02:00 2m

20.5 Without disturbing the pellet, transfer 200 µL of the the supernatant (clarified lysate) to the Sample Plate containing the Bead/PK Mix from Step 15 and proceed to Step 22.

21

Processing samples in Plates

21.1 Add 454 µL of the Lysis Solution to the appropriate wells of a deep-well plate.

21.2 Transfer 200 µL of the supernatant of the fecal slurry to the tube.

21.3 Seal the plate with sealing foil.

21.4 Shake the plate at moderate speed for 5 minutes.

21.5 Centrifuge at 3000 x g, Room temperature, 00:05:00 . 5m

21.6 Remove the supernatant (clarified lysate) without disturbing the pellet and transfer 200 µL to the Sample Plate containing the Bead/PK Mix from Step 15 and proceed to Step 22.

22 Mix the clarified lysate (sample) and bead mix for 00:02:00 at Room temperature . 2m

23 Add 350 µL of MagMAX™ CORE Binding Solution to the clarified lysate-bead mixture in wells of the Sample Plate.

24 Immediately proceed to process the samples on the KingFisher™ Flex instrument

25 Select the appropriate script on the instrument. **KingFisher Flex heated script: MagMAX CORE_Flex.bdz** 27m

00:27:00

> Note
>
> Processing on the machine takes approximately 27 minutes.

PCR Method - Setting up and running the PCR reactions

26 The probe is labeled with FAM with Black Hole Quencher 1 (BHQ1) so detection parameters appropriate for FAM or SYBR should be used.The developing laboratory uses Applied Biosystems™ VetMAX™ Xeno™ Internal Positive Control DNA (ThermoFisher Scientific A29762 or A29764) and the VetMAX™ Xeno™ Internal Positive Control - VIC™ Assay (ThermoFisher Scientific A29765 or A29767) to monitor for PCR inhibition.

PCR Method - DNA Extraction

1m

27 After samples have finished processing, proceed with PCR or store samples at 4 °C until ready to perform PCR.

PCR Method - Setting up and running the PCR reactions

28 Create run on ABI 7500 Fast machine. Setting up the ABI 7500 Fast Thermocycler: Refer to the manual provided for running the machine. Reaction times are as follows in Table 1.

Equipment

7500 Fast Real-Time PCR System, desktop	NAME
Applied Biosystems	BRAND
4351107	SKU

	A	B	C	D	E	F
	Stage	**Function**	**Temperature**	**Duration (sec)**	**Repeats**	**Optics**
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 4a. PCR amplification conditions in ABI 7500 under standard conditions are as follows:

	A	B	C	D	E	F
	Stage	**Function**	**Temperature**	**Duration (sec)**	**Repeats**	**Optics**
		Reverse Transcription	50°C	300		
	Stage 1	Initial PCR Activation	95.0°C	20		Off

	A	B	C	D	E	F
	Stage 2	Annealing	95.0°C	3	Repeats 45 times	Off
	Stage 2	Extension	60°C	60		On

Table 4b. PCR amplification conditions in ABI 7500 under FAST conditions are as follows:

Note

Passive reference dye should be set as ROX

29 Calculate the number of PCR reactions to be performed. Using Table 5 as a guide, determine the quantity of each reagent needed. Include a positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560) and negative (*E. coli* ATCC 25922) control and a "No Template" negative control.

For calculations, reference **Table 5.** PCR Worksheet for Reagent Volumes for Master Mix Preparation for *Campylobacter jejuni* PCR.

Table 5. PCR Worksheet for Reage...

30 Combine reagents except for the DNA template in a 1.5ml or 2ml microcentrifuge tube and mix well.

31 Aliquot 18 µL into each PCR tube or reaction well if using a plate.

32 Add samples and positive (*C. jejuni* ATCC 33291, ATCC 33292, or ATCC 33560 DNA) and negative (*E. coli* ATCC 25922 DNA) controls along with PCR inhibition controls

VetMAX Xeno Internal Positive Control ND **Thermo Fisher Scientific Catalog** #A29762/A29764

33 Place Samples in thermocycler and perform run.

34 Following run, analyze the results. Results should be reported with a Ct value and interpretation.

35 Samples with a Ct value of 40 or less should be interpreted as positive. Samples with a Ct value >40 or an undetermined Ct value should be considered negative.

	A	B	C
	Target	**CT Value**	**Interpretation**
	gyrA	<=40	Detected
		>40	Not detected

Table 6.Interpretation of Ct Value

	A	B
	Oligo Name	**5′-3′ Sequence**
	gyrA-F	AAGATACGGTCGATTTTGTTCCA
	gyrA-R	CTACAGCTATACCACTTGAACCATTTAATA
	gyrA-P	5′-[FAM]TGATGGTTCAGAAAGCGAACCTGATGTTTT[BHQ1] 3′

Table 7. Primers and Probe

Protocol references

Life Technologies Corporation. 2017. MagMAX™ CORE Mechanical Lysis Module USER GUIDE. Publication Number MAN0015945 Revision A.0. Austin, TX. https://www.thermofisher.com/document-connect/document-connect.html?url=https://assets.thermofisher.com/TFS-Assets%2FLSG%2Fmanuals%2FMAN0015945-MagMAXCORE-Mech-Lysis-Module-UG.pdf